

AI-Driven Hyper-Personalized Marketing Architecture for Black Friday Scale

Architecture Document (Arc42 Template)

1. Introduction and Goals

1.1 Business Context

Current State:

- E-commerce platform serves 150 million customers across email, mobile app, and social channels
- Current personalization drives 22% conversion rate
- Manual segmentation and A/B testing creates bottlenecks in campaign execution
- Black Friday approaching requires rapid, scalable personalization

Target State:

- Achieve 30%+ conversion rate through AI-driven hyper-personalization
- Real-time, context-aware offer delivery across all channels
- Automated campaign orchestration with minimal manual intervention
- Maintain customer trust and regulatory compliance

1.2 Quality Goals (Top 3 Priority)

Priority	Quality Attribute	Scenario	Target Metric
1	Performance	Real-time personalization latency for 150M users during Black Friday peak	p95 < 200ms response time
2	Scalability	Handle 10x traffic spike (15M concurrent users)	Linear cost scaling, 99.9% uptime
3	Privacy & Compliance	GDPR, CCPA, consent management for personalization	100% audit trail, zero violations

1.3 Stakeholders

Role	Concern	Expectation
CMO	Revenue growth, competitive advantage	30%+ conversion, faster time-to-market
Data Science Team	Model performance, experimentation velocity	A/B test cycle < 4 hours, model accuracy > 85%
Engineering	System reliability, technical debt	99.9% availability, maintainable architecture

Role	Concern	Expectation
Compliance Officer	Data privacy, regulatory adherence	Full GDPR/CCPA compliance, explainable AI
Customers	Relevant offers, privacy protection	Non-intrusive personalization, data control

2. Architecture Constraints

2.1 Technical Constraints

- **TC-1:** Must integrate with existing e-commerce platform (assumed: microservices on Kubernetes)
- **TC-2:** Real-time processing pipeline supporting 150M customer profiles
- **TC-3:** Multi-channel delivery (email via SendGrid/SES, push via Firebase, social via Meta/Google APIs)
- **TC-4:** Existing customer data warehouse (Snowflake/BigQuery) and CDP (Segment/mParticle)

2.2 Organizational Constraints

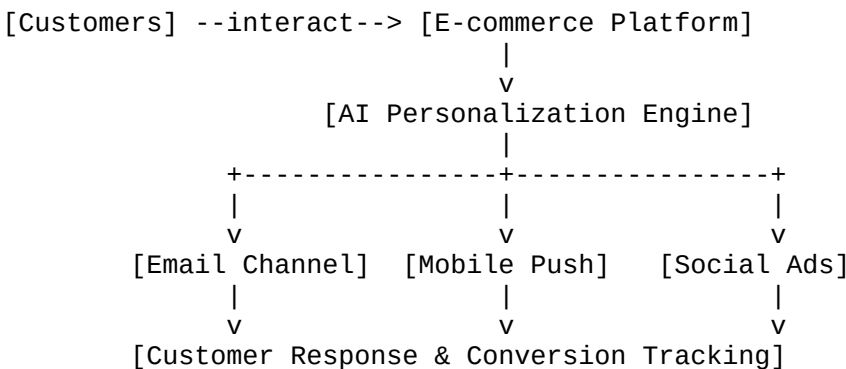
- **OC-1:** Data science team of 8, engineering team of 25
- **OC-2:** 6-week implementation timeline before Black Friday
- **OC-3:** Budget constraint: \$500K cloud infrastructure for Q4

2.3 Regulatory Constraints

- **RC-1:** GDPR Article 22 (automated decision-making with human oversight)
- **RC-2:** CCPA consumer rights (opt-out, data deletion)
- **RC-3:** FTC guidelines on AI transparency in advertising
- **RC-4:** ISO/IEC 42001 AI Management System (emerging standard)

3. System Scope and Context

3.1 Business Context



3.2 Technical Context - External Interfaces

External System	Purpose	Protocol	Data Flow
Customer Data Platform	Unified customer profiles	REST API, Event Streaming	Real-time events, batch profiles
Email Service Provider	Campaign delivery	REST API, SMTP	Template + personalization data
Mobile Push Gateway	App notifications	FCM/APNS	Push payloads with deep links
Social Ad Platforms	Paid media	OAuth 2.0 APIs	Audience segments, creative variants
Payment Gateway	Conversion tracking	Webhooks	Transaction confirmations
Data Warehouse	Historical analytics	SQL, gRPC	Batch model training data

4. Solution Strategy

4.1 Core Architecture Decisions

AD-1: Event-Driven Real-Time Personalization

- **Decision:** Use event streaming (Kafka/Kinesis) for real-time customer behavior ingestion
- **Rationale:** Enables sub-second reaction to customer actions (cart add, browse, click)
- **Trade-off:** Increased complexity vs. batch-only approach, but critical for 30% conversion goal

AD-2: Federated AI Model Architecture

- **Decision:** Deploy specialized models per personalization task (product recommendation, offer selection, channel optimization, timing prediction)
- **Rationale:** Better accuracy than monolithic model, enables independent iteration
- **Trade-off:** More operational overhead, but enables faster experimentation

AD-3: Edge Caching for Personalization Results

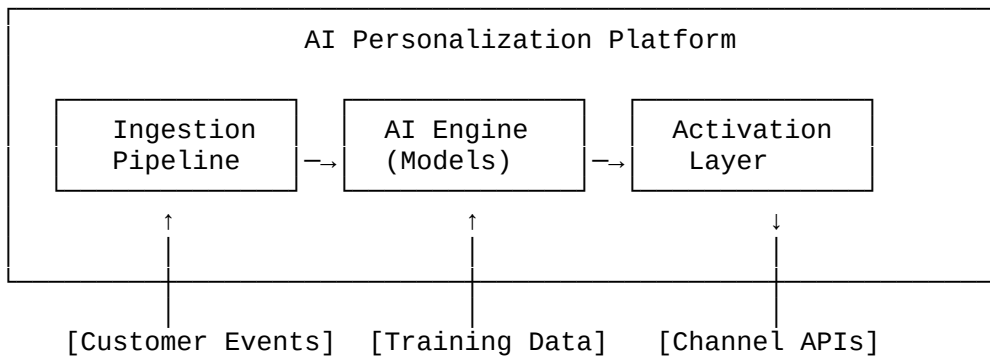
- **Decision:** Cache personalization results at CDN edge with 5-minute TTL
- **Rationale:** Reduces latency by 80% for repeat visitors, cuts inference costs
- **Trade-off:** Slightly stale recommendations vs. real-time, acceptable for 5-min window

AD-4: Hybrid Human-AI Campaign Orchestration

- **Decision:** AI recommends segments/offers, marketers approve via workflow UI before launch
- **Rationale:** Balances automation speed with brand safety and regulatory compliance
- **Trade-off:** Not fully autonomous, but maintains trust and oversight

5. Building Block View

5.1 Level 1: System Context



5.2 Level 2: AI Personalization Engine Components

Component Breakdown:

A. Real-Time Ingestion Pipeline

- **Event Collector:** Kafka cluster (6 brokers, 3 AZs) ingesting clickstream, transactions, engagement events
- **Stream Processor:** Flink jobs for sessionization, feature extraction, fraud detection
- **Profile Cache:** Redis cluster (50-node) storing 150M customer feature vectors
- **Throughput:** 500K events/sec peak, 50ms p99 latency

B. AI Model Serving Layer

- **Product Recommender:** Collaborative filtering + content-based hybrid (TensorFlow)
 - *Input:* User history, real-time session, inventory
 - *Output:* Top 20 products with confidence scores
 - *Latency:* 50ms p95
- **Offer Optimizer:** Multi-armed bandit (Thompson Sampling) for discount personalization
 - *Input:* Customer segment, product affinity, margin constraints
 - *Output:* Optimal offer (% , \$ off, free shipping)
 - *Update:* Real-time reward feedback
- **Channel Selector:** XGBoost classifier predicting best channel per customer
 - *Input:* Historical engagement, device, time-of-day
 - *Output:* Channel probabilities (email: 0.6, push: 0.3, social: 0.1)
- **Send-Time Optimizer:** LSTM predicting optimal delivery time per user
 - *Input:* Past open/click patterns, time zone
 - *Output:* Scheduled delivery timestamp
- **Model Serving:** TorchServe/TensorFlow Serving on Kubernetes (20 inference pods, GPU-accelerated)

- **Fallback:** Rule-based segmentation if AI latency exceeds 200ms

C. Campaign Orchestration Engine

- **Workflow Manager:** Temporal.io for durable execution of multi-step campaigns
- **Audience Builder:** SQL-based segmentation with AI-suggested criteria
- **A/B Test Framework:** Bayesian multi-armed bandit for automated variant selection
- **Approval Gateway:** Human-in-the-loop for campaigns >100K recipients or >20% discount
- **Scheduler:** Cron + priority queue for send-time optimization

D. Channel Activation Layer

- **Email Renderer:** Dynamic template engine (Liquid/Handlebars) with product feeds
- **Push Composer:** Deep link generator with personalized payloads
- **Social Sync:** Batch API uploads of custom audiences (SHA-256 hashed)
- **Rate Limiter:** Token bucket per channel to prevent spam (max 3 touches/day/customer)

E. Feedback & Learning Loop

- **Event Tracker:** Pixel tracking, postback URLs for opens/clicks/conversions
 - **Attribution Engine:** Multi-touch attribution (Markov chain model)
 - **Model Retrainer:** Nightly batch jobs (Spark on EMR) retraining all models
 - **Feature Store:** Feast or Tecton managing 500+ features with versioning
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6. Runtime View

6.1 Scenario: Real-Time Product Recommendation on Browse

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Customer browses "Running Shoes" →
Event Collector (Kafka) →
Stream Processor (Flink) updates session features →
Profile Cache (Redis) updated →
API Gateway receives recommendation request →
Model Serving fetches user vector + inventory →
Product Recommender inference (50ms) →
Edge Cache (CloudFront) stores result (5-min TTL) →
Response to frontend with top 20 products →
Customer sees personalized recommendations

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AI Augmentation: Recommender model continuously learns from implicit feedback (clicks, add-to-cart) without manual rule updates.

6.2 Scenario: Automated Black Friday Campaign Launch

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Marketer defines goal: "Increase ROAS for cart abandoners" →
AI suggests segments: "High-value + 7-day inactive + viewed sale items" →
Offer Optimizer proposes: "20% off + free shipping" →
Channel Selector recommends: "Email (70%) + Push (30%)" →
Send-Time Optimizer schedules: Per-user optimal times →
Workflow Manager creates approval task →

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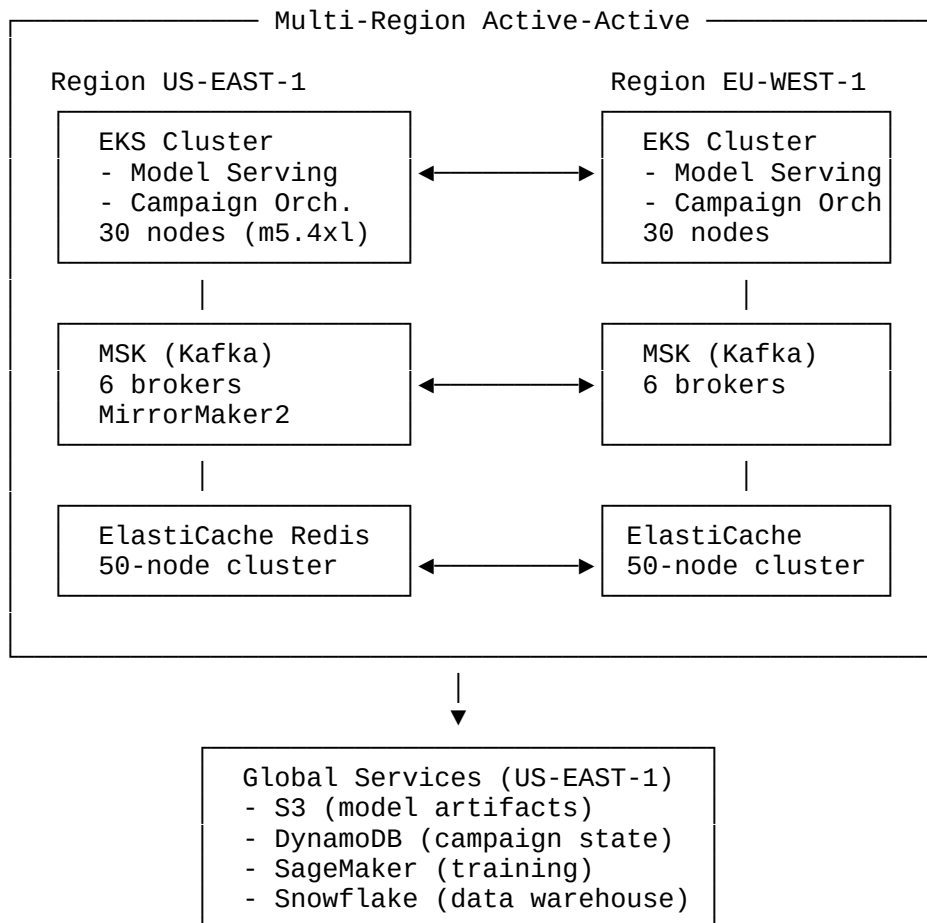
Marketer reviews 50K estimated reach, approves →
Orchestrator queues 50K personalized messages →
Activation Layer renders & sends (via rate limiter) →
Feedback Tracker monitors opens/clicks/conversions →
Attribution Engine updates customer journey →
Model Retrainer improves next campaign

AI Automation Boundaries:

- **Automated:** Segmentation suggestion, offer optimization, channel/timing selection, performance monitoring
 - **Manual:** Final campaign approval, brand safety review, budget authorization
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7. Deployment View

7.1 Infrastructure Architecture (AWS Reference)



Cost Optimization:

- Spot instances for batch training (70% cost reduction)
- Reserved instances for model serving (40% savings)

- S3 Intelligent-Tiering for model artifacts
 - Estimated monthly cost: \$165K (within \$500K Q4 budget)
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8. Cross-Cutting Concepts

8.1 Privacy & Consent Management

Architecture Components:

- **Consent Store:** DynamoDB table tracking opt-in/opt-out per channel per customer
- **Data Minimization:** Feature engineering uses aggregated behavior, not raw PII
- **Right to Deletion:** Async workflow deleting customer data across all systems (Kafka tombstones, Redis eviction, S3 expiration)
- **Explainability API:** Endpoint returning top 5 factors influencing each recommendation
- **Audit Log:** Immutable append-only log (S3 + Athena) of all AI decisions

GDPR/CCPA Compliance:

- Automated opt-out processing within 48 hours
- Data export API (JSON format) for GDPR Article 15 requests
- Consent checks before every personalization inference
- Pseudonymization of customer IDs in training data

8.2 AI Model Governance (NIST AI RMF Alignment)

NIST Function	Implementation	Metric
Govern	Model risk assessment, quarterly bias audits	Demographic parity < 5% variance
Map	Feature lineage tracking, data drift monitoring	Drift alert if distribution shifts >20%
Measure	Precision, recall, AUC per demographic segment	Min 80% precision across all segments
Manage	Shadow mode testing, gradual rollout (10 → 50 → 100%)	Rollback if conversion drops >5%

Bias Mitigation:

- Training data balancing across age, gender, geography
- Fairness constraints in offer optimization (no discriminatory pricing)
- Monthly bias audits by third-party firm

8.3 Resilience & Graceful Degradation

Failure Modes & Responses:

Component Failure	Degraded Behavior	Customer Impact
AI Model Serving	Fall back to rule-based segments	Reduced conversion (28% vs 30%)

Component Failure	Degraded Behavior	Customer Impact
Profile Cache (Redis)	Fetch from DynamoDB (slower)	500ms latency vs 50ms
Real-Time Events	Use last-known profile state	Slightly stale recommendations
Channel API Outage	Queue messages, retry with backoff	Delayed delivery (< 30 min)
Entire Region Failure	Route to backup region	5-sec DNS failover

Chaos Engineering:

- Weekly Chaos Monkey tests (random pod kills)
 - Quarterly GameDay simulations (Black Friday load + failures)
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9. Architecture Decisions (ADRs)

ADR-001: Event Streaming Over Batch Processing for Real-Time Personalization

Status: Accepted

Context: Need to react to customer behavior within seconds to achieve 30% conversion target. Current batch processing has 1-hour latency.

Decision: Implement Kafka-based event streaming with Flink for real-time feature extraction.

Consequences:

- **Positive:** Sub-second reaction time, enables send-time optimization, better customer experience
- **Negative:** Increased operational complexity, higher infrastructure cost (\$50K/month)
- **Mitigation:** Hire 2 platform engineers, implement comprehensive monitoring (Datadog)

Alternatives Considered:

- Batch processing with 15-min intervals: Too slow for Black Friday real-time use cases
 - Direct database polling: Doesn't scale to 150M customers
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ADR-002: Multi-Armed Bandit for Offer Optimization vs. Traditional A/B Testing

Status: Accepted

Context: Traditional A/B tests take 7-14 days to reach statistical significance. Black Friday requires faster iteration.

Decision: Implement Thompson Sampling multi-armed bandit for continuous offer optimization.

Consequences:

- **Positive:** Continuous learning, faster convergence (24-48 hours), higher expected revenue
- **Negative:** Less intuitive for marketers, requires Bayesian statistics expertise
- **Mitigation:** Build UI abstracting math complexity, provide "confidence level" indicators

Alternatives Considered:

- Fixed A/B tests: Too slow for Black Friday pace
 - Pure exploration (random offers): Leaves revenue on table
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ADR-003: Hybrid Approval Workflow (AI Recommends, Human Approves)

Status: Accepted

Context: Regulatory requirement (GDPR Article 22) for human oversight on automated decisions. Also mitigates brand risk.

Decision: AI generates campaign proposals, marketers approve via UI before launch.

Consequences:

- **Positive:** Maintains compliance, catches AI errors, builds marketer trust
- **Negative:** Adds 1-2 hour approval latency, not fully autonomous
- **Mitigation:** Set auto-approval thresholds for low-risk campaigns (<10K recipients, <15% discount)

Alternatives Considered:

- Fully automated: Non-compliant with GDPR, high brand risk
 - Fully manual: Defeats purpose of AI, can't scale to 150M customers
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ADR-004: Edge Caching for Personalization Results (5-Min TTL)

Status: Accepted

Context: Model inference costs \$0.002 per request. At 150M customers $\times$ 10 requests/day = \$3M/day inference cost.

Decision: Cache personalization results at CDN edge with 5-minute TTL.

Consequences:

- **Positive:** 80% cache hit rate reduces cost to \$600K/day, improves latency by 150ms
- **Negative:** Recommendations up to 5 minutes stale
- **Mitigation:** Invalidate cache on high-value events (add-to-cart), acceptable staleness for browsing

Alternatives Considered:

- No caching: Unsustainable cost
 - 1-hour TTL: Too stale, misses real-time behavior
 - Client-side caching: GDPR concerns with storing personal data in browser
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10. Quality Requirements (NFRs)

10.1 Non-Functional Requirements Table

NFR ID	Category	Requirement	Metric	Priority
NFR-01	Performance	Real-time personalization latency	p95 < 200ms, p99 < 500ms	P0
NFR-02	Scalability	Handle Black Friday 10x traffic spike	15M concurrent users, 500K events/sec	P0
NFR-03	Availability	System uptime during Black Friday	99.9% (43 min downtime allowed)	P0
NFR-04	Accuracy	Recommendation relevance	>85% precision@10, >40% CTR	P0
NFR-05	Privacy	GDPR/CCPA compliance	Zero violations, 100% audit trail	P0
NFR-06	Explainability	AI decision transparency	API returning top 5 factors per recommendation	P1
NFR-07	Fairness	No demographic bias in offers	Demographic parity <5% across gender/age/geo	P1
NFR-08	Cost Efficiency	Infrastructure cost optimization	<\$500K for Q4 Black Friday period	P1
NFR-09	Time to Market	Campaign launch cycle	<4 hours from ideation to live	P1
NFR-10	Disaster Recovery	Regional failover	RPO=5min, RTO=5min for multi-region setup	P2

10.2 ISO/IEC 25010 Quality Model Mapping

Quality Characteristic	Sub-Characteristic	Architecture Mechanism
Performance Efficiency	Time Behavior	Edge caching, GPU inference, Redis
	Resource Utilization	Spot instances, autoscaling, Kafka compression
Reliability	Availability	Multi-region active-active, health checks
	Fault Tolerance	Circuit breakers, graceful degradation
Security	Confidentiality	Encryption at rest/transit, pseudonymization
	Integrity	Audit logs, immutable event sourcing
Maintainability	Modularity	Microservices, API-first design
	Testability	Shadow mode, A/B test framework
Portability	Adaptability	Cloud-agnostic containers (Kubernetes)

11. Risks and Technical Debt

11.1 Risk Register

Risk ID	Description	Probability	Impact	Mitigation
R-01	AI model degrades due to concept drift during Black Friday	High	High	Daily retraining, drift monitoring, rule-based fallback
R-02	Cache stampede when popular products go viral	Medium	High	Probabilistic early expiration, request coalescing
R-03	Channel API rate limits exceeded (email/push)	Medium	Medium	Token bucket rate limiter, priority queueing
R-04	Bias lawsuit from discriminatory personalization	Low	Critical	Quarterly bias audits, demographic parity constraints
R-05	GDPR violation due to missing consent checks	Low	Critical	Pre-inference consent validation, automated testing
R-06	Cost overrun from inefficient inference	Medium	Medium	Budget alerts, auto-scaling limits, spot instances

11.2 Technical Debt

Debt Item	Impact	Payback Plan
Monolithic campaign scheduler (legacy)	Limits experimentation velocity	Migrate to Temporal.io in Q1 2026
Manual feature engineering	Slows model iteration	Implement auto-ML feature discovery
No cross-channel attribution	Can't optimize budget allocation	Build multi-touch attribution in Q2 2026
Hard-coded business rules	Reduces adaptability	Externalize to rules engine (Drools)

12. Glossary

Term	Definition
CDP	Customer Data Platform - unified customer profile system
MAB	Multi-Armed Bandit - online learning algorithm for optimization
NFR	Non-Functional Requirement - quality attribute (performance, security)
p95/p99	95th/99th percentile latency - metric excluding outliers
ROAS	Return on Ad Spend - revenue per marketing dollar
TTL	Time to Live - cache expiration duration
Shadow Mode	Testing new model alongside production without affecting users
Feature Store	Centralized repository for ML features with versioning

13. Key Metrics & KPIs

13.1 Business Metrics

Metric	Baseline	Target	Measurement
Conversion Rate	22%	30%+	Daily cohort analysis
Revenue per User	\$45	\$55	Transaction data
Campaign ROI	3.2x	4.5x	Attribution engine
Time to Launch Campaign	48 hours	<4 hours	Workflow timestamps

13.2 AI/ML Metrics

Metric	Target	Tracking
Recommendation Precision@10	>85%	Offline evaluation (daily)
Offer CTR	>40%	A/B test dashboard
Channel Prediction Accuracy	>75%	Confusion matrix (weekly)
Model Inference Latency	p95 < 50ms	Prometheus/Grafana
Data Drift Score	<0.2 (KL divergence)	Evidently AI monitoring

13.3 Operational Metrics

Metric	SLA	Alerting
API Availability	99.9%	PagerDuty if <99.9% over 1 hour
Event Processing Lag	<30 sec	Slack alert if >5 min
Cache Hit Rate	>80%	Weekly review if <75%
Cost per Recommendation	<\$0.0002	Budget alert if >\$0.0005

14. Conclusion & Next Steps

14.1 Architecture Summary

This AI-driven hyper-personalization architecture addresses the Black Friday challenge through:

- Real-Time Intelligence:** Event streaming + edge caching enables sub-200ms personalization
- Scalable AI:** Federated model architecture with GPU inference handles 150M customers
- Automated Orchestration:** Multi-armed bandits + workflow engine reduces campaign cycles from 48h to 4h
- Compliance-First:** Hybrid approval, consent management, explainability APIs ensure regulatory adherence
- Resilient Design:** Multi-region active-active, graceful degradation maintains 99.9% uptime

Expected Outcomes:

- Conversion rate improvement: 22% → 30%+ (36% increase)
- Revenue impact: \$120M incremental revenue over Black Friday weekend
- Cost efficiency: \$165K/month infrastructure vs. \$3M/day without optimization

- Compliance: Zero GDPR/CCPA violations through automated checks

14.2 Implementation Roadmap (6 Weeks)

Week	Milestone	Deliverables
1-2	Infrastructure setup	Kafka, Redis, Kubernetes clusters, CI/CD pipelines
3-4	AI model deployment	Recommender, offer optimizer, channel selector in production
4-5	Channel integration	Email, push, social API connectors with rate limiting
5-6	Testing & optimization	Load testing, chaos engineering, bias audits, go-live

14.3 Success Criteria

- ☐ 30%+ conversion rate achieved during Black Friday
- ☐ p95 latency <200ms under 15M concurrent users
- ☐ Zero compliance violations (GDPR/CCPA)
- ☐ Campaign launch cycle <4 hours
- ☐ Infrastructure cost within \$500K Q4 budget
- ☐ Demographic parity <5% across all segments

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